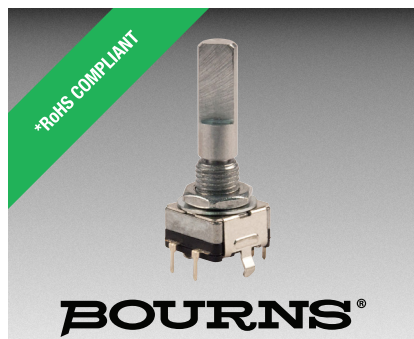




Features

- Push switch option
- High detent torque
- High rotational life and switch life
- Metal bushing/shaft
- RoHS compliant*



PEC11H Series - 11 mm Encoder w/High Detent Force

Electrical Characteristics

Output.....	2-bit quadrature code
Closed Circuit Resistance	100 milliohms maximum
Contact Rating.....	10 mA @ 5 VDC
Insulation Resistance	100 megohms @ 250 VDC
Dielectric Withstanding Voltage - Sea Level.....	300 VAC minimum
Electrical Travel	Continuous
Contact Bounce (60 RPM).....	3.0 ms maximum**
RPM (Operating)	60 maximum**

Environmental Characteristics

Operating Temperature Range	-20 °C to +70 °C (-4 °F to +158 °F)
Storage Temperature Range	-35 °C to +85 °C (-31 °F to +185 °F)
Humidity.....	MIL-STD-202, Method 103B, Condition A
Vibration	10~55~10 Hz / 1 min. / Amplitude 1.5 mm
Shock.....	100 G
Rotational Life.....	100,000 cycles minimum
IP Rating.....	IP 40
Moisture Sensitivity Level	1

Mechanical Characteristics

Mechanical Angle	360 ° continuous
Torque	
Detent	210 ±70 gf-cm (2.0 ± 0.1 oz-in)
Mounting	70 N-cm (6.2 lb-in) maximum
Shaft Strength (Push-Pull).....	50 N (11.2 lbs)
Weight	5 gm (0.17 oz) maximum
Terminals	Printed circuit board terminals
Soldering Condition	
Wave Soldering	Sn95.5/Ag2.8/Cu0.7 solder with no-clean flux: 260 °C ±5 °C max. for 3 ± 1 seconds
Hand Soldering.....	Not recommended
Hardware	One flat washer and one mounting nut supplied with each encoder

Switch Characteristics

Switch Type	Contact Push ON Momentary SPST
Switch Life	30,000 actuations minimum
Power Rating (Resistive Load)	10 mA at 5 V DC
Switch Travel	0.5 ± 0.3 mm
Switch Actuation Force	610 ± 306 gf (8.47 ± 4.24 oz.-in.)

Additional Information

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[SAMPLES](#)



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WARNING

Cancer and Reproductive Harm

www.P65Warnings.ca.gov

*RoHS Directive 2015/863, Mar 31, 2015 and Annex.

**Devices are tested using standard noise reduction filters. For optimum performance, designers should use noise reduction filters in their circuits. Specifications are subject to change without notice. Users should verify actual device performance in their specific applications. The products described herein and this document are subject to specific legal disclaimers as set forth on the last page of this document, and at www.bourns.com/docs/legal/disclaimer.pdf.



CAUTION

Do not store product in high temperature and humidity, direct sunlight and/or places where corrosive gases may be generated. Please use product within 6 months from the date of delivery and promptly after unpacking.

Applications

Level control, tuning and timer settings in:

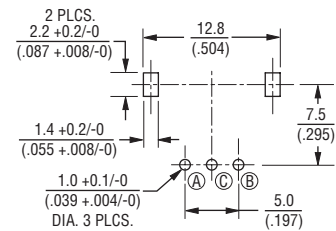
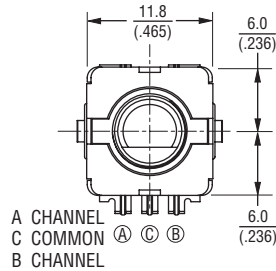
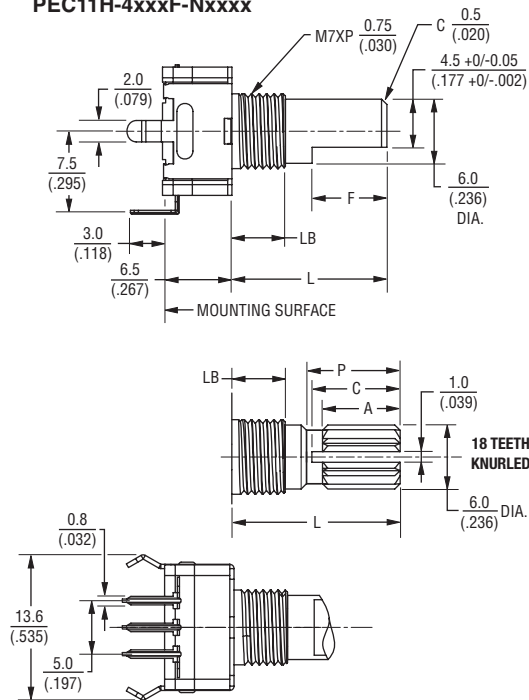
- Audio-visual equipment
- Consumer electric appliances
- Environmental controls
- Musical instrumentation
- Communications equipment

PEC11H Series - 11 mm Encoder w/High Detent Force

BOURNS®

Product Dimensions

PEC11H-4xxxF-Nxxxx

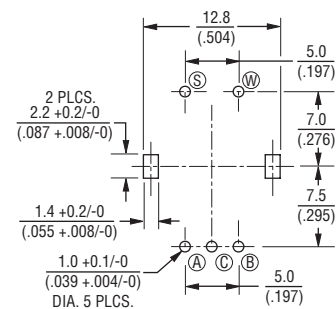
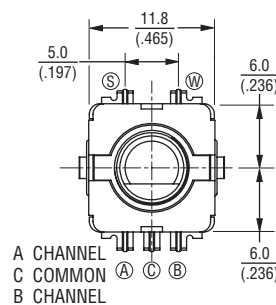
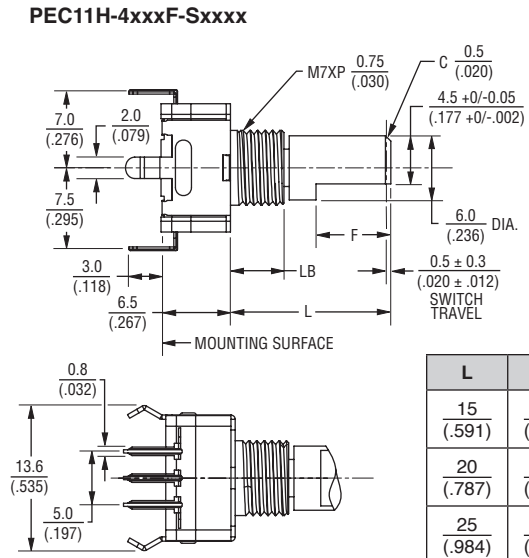


L	LB	F	P	A	C
15 (.591)	5.0 (.197)	5.0 (.197)	7.0 (.276)	6.0 (.236)	6.0 (.236)
20 (.787)	7.0 (.276)	10.0 (.394)	7.0 (.276)	6.0 (.236)	7.0 (.276)
25 (.984)	7.0 (.276)	12.0 (.472)	12.0 (.472)	10.0 (.394)	11.0 (.433)

Dim.	Tol.
<10.0 (<.394)	±0.3 (±.012)
10.0 ~ 100 (.394 ~ 3.937)	±0.5 (±.020)

DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

PEC11H-4xxxF-Sxxxx



L	LB	F	Dim.	Tol.
15 (.591)	5.0 (.197)	7.0 (.276)	<10.0 (<.394)	±0.3 (±.012)
20 (.787)	7.0 (.276)	10.0 (.394)	10.0 ~ 100 (.394 ~ 3.937)	±0.5 (±.020)
25 (.984)	7.0 (.276)	12.0 (.472)		

DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

Specifications are subject to change without notice.

Users should verify actual device performance in their specific applications.

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PEC11H Series - 11 mm Encoder w/High Detent Force

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How To Order

Model **PEC11H - 4 0 20 F - S 0020**

Terminal Configuration _____
4 = PC Pin Horizontal/Rear Facing

Detent Option _____
0 = 16 Detents (8, 16 pulses) 2 = 24 Detents (24 pulses)
1 = 20 Detents (20 pulses) 3 = 30 Detents (15 pulses)

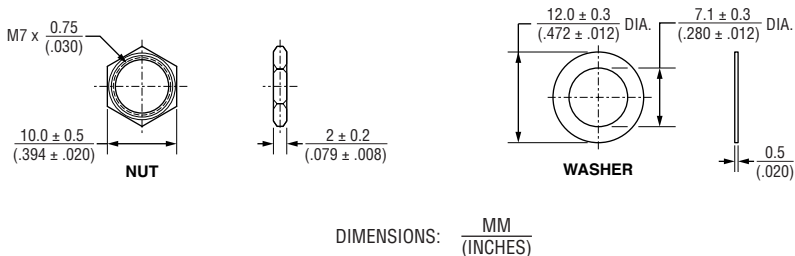
Standard Shaft Length _____
15 = 15.0 mm 25 = 25.0 mm
20 = 20.0 mm 30 = 30.0 mm

Shaft Style _____
F = Metal Flatted Shaft
K = Metal Knurled Shaft

Switch Configuration _____
S = Push Momentary Switch
N = No Switch

Resolution _____
0008 = 8 Pulses per 360 ° Rotation 0020 = 20 Pulses per 360 ° Rotation
0015 = 15 Pulses per 360 ° Rotation 0024 = 24 Pulses per 360 ° Rotation
0016 = 16 Pulses per 360 ° Rotation

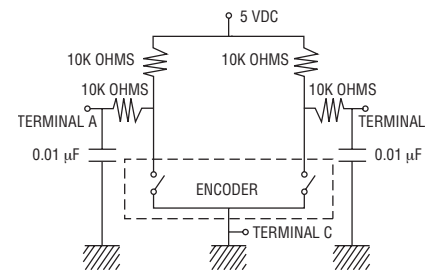
Hardware



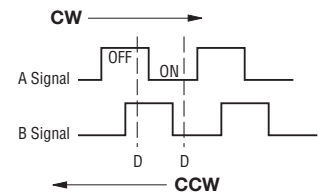
Switch Circuit



Suggested Filter Circuit



Quadrature Output Table



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